FABRICFLOW RESEARCH REPORT

Component 3

Dataset & ML Model Results

Emergency Situation Detection and Management

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Generated
2026-10-04

EXECUTIVE SUMMARY

- 598 real historical daily records across 11 independent bulk orders form the evidence base.
- Random Forest was selected for current risk type and overall order-risk classification.
- Three calibrated next-3-production-day subtype warnings are supported: machine breakdown, quality limit, and output/schedule risk.
- Evaluation uses Leave-One-Group-Out validation by Bulk Order ID; augmented rows are used only inside training folds.
- Artifacts remain research-only because real-order coverage and minority-class evidence are limited.

Evidence source: current repository reports and metadata. No metrics are estimated or invented.

Dataset inventory and provenance

What data exists, what was used, and how synthetic rows were controlled

HISTORICAL ROWS
598
Real daily records retained for validation

INDEPENDENT ORDERS
11
Bulk Order ID is the validation group

STYLES
11
One style per historical order

ELIGIBLE WARNING ROWS
174
Stable days with a full 3-day horizon

Dataset	Size	Training additions	Role
component3_final_preprocessed_dataset.xlsx	598 real rows	11 sheets / orders	Primary historical evidence
component3_training_augmented.xlsx	1,962 physical rows	479 new augmented rows	Risk-type training; real-only validation
component3_training_balanced_research_v2.xlsx	2,208 physical rows	725 augmented training rows	Overall-risk training; real-only validation

Real risk-type class coverage

Risk class	Real rows	Orders containing c
No Issue	193	11
Minor Delay	4	1
Working Hours Issue	128	10
Worker Issue	188	11
Commitment Too Low	37	9
Machine Breakdown Issue	44	11
Quality Issue	4	4

Data-quality controls

No duplicate order-date rows, no cumulative-output mismatch rows, and no cumulative totals above full-order quantity were reported. Generated rows are identified and excluded from validation.

Important limitation

Minor Delay and Quality Issue have only four real rows each. Model scores for these rare classes must not be interpreted as strong production evidence.

Sources: data/*.xlsx; reports/component3_v2; reports/component3_model2_v3

Data preparation and validation design

Order-level isolation prevents records from the same order leaking across train and test

VALIDATION PIPELINE

1 **Group by order**
Bulk_Order_ID defines independent groups

→

2 **Hold out one order**
One unseen order becomes the test fold

→

3 **Fit preprocessing**
Transforms are fitted on training folds only

→

4 **Add augmentation**
Synthetic rows enter training folds only

→

5 **Predict held-out order**
Repeat for all 11 orders and combine scores

Why this is stronger than a random row split

Daily rows from the same bulk order are correlated. A random 80/20 split could expose the test set to the same order seen in training. LOGO instead measures transfer to an unseen order.

Early-warning calibration

Calibrated warning artifacts use nested grouped validation. A sigmoid calibrator and each target-specific decision threshold are learned without using the held-out order.

Feature groups

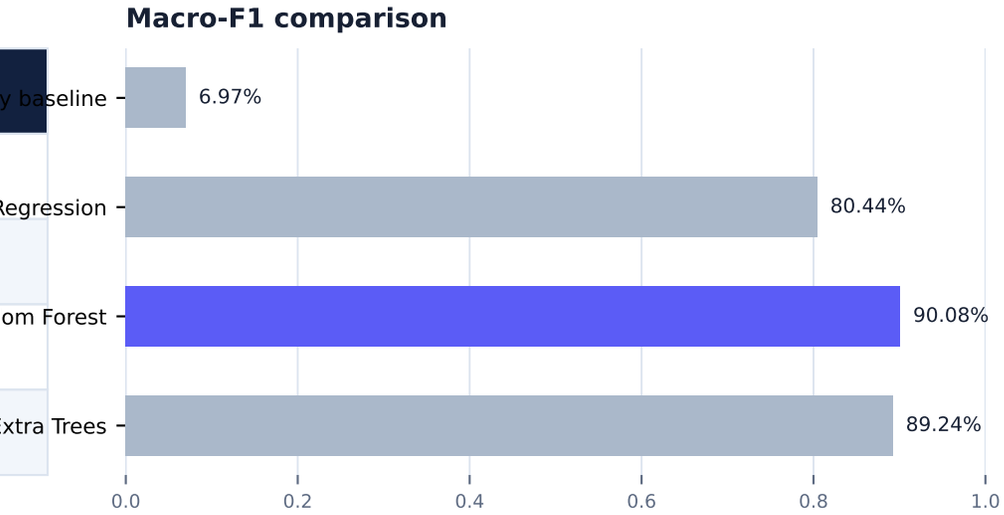
Group	Examples
Production	Daily commitment, actual output, output gap, cumulative completed
Disruption	Machine breakdowns, worker shortages, damage quantity and ratio
Schedule	Working day, days remaining, progress %, required daily rate
Process	Cutting/sewing phase indicators and order quantities
Early-warning history	Previous output, trailing 3-day averages/variation, recent emergency counts

Leakage exclusions include IDs, dates, final outcomes, risk labels, recommendations, completion date, and future warning labels.

Model 1 — current risk type

Seven-class classification using real-order LOGO validation

Candidate	Accuracy	Macro-F1	Weighted-F1	Decision
Dummy baseline	32.27%	6.97%	15.75%	
Logistic Regression	76.42%	80.44%	76.94%	
Random Forest	81.77%	90.08%	81.93%	Selected
Extra Trees	80.10%	89.24%	80.17%	



SELECTED MODEL

Random Forest

Best unseen-order Macro-F1

ACCURACY

81.77%

598 real validation rows

MACRO-F1

90.08%

Equal weight across seven classes

WEIGHTED-F1

81.93%

Weighted by class support

What the model predicts

The current production-day condition: No Issue, Minor Delay, Working Hours Issue, Worker Issue, Commitment Too Low, Machine Breakdown Issue, or Quality Issue.

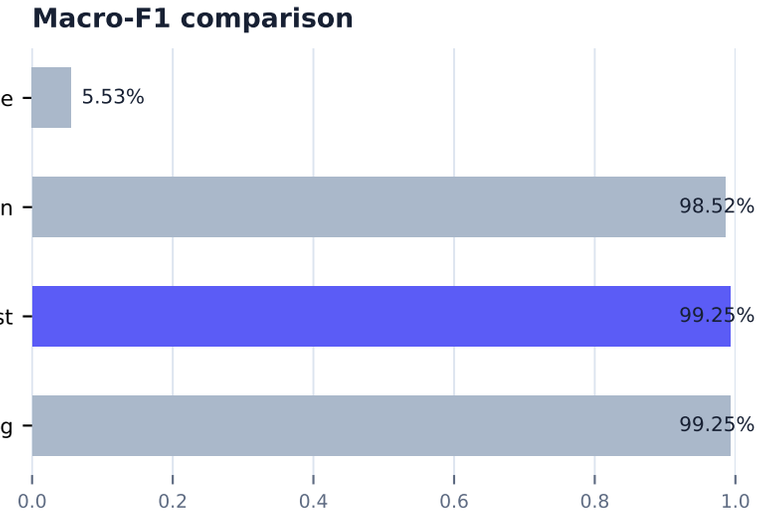
Observed error pattern

Most remaining confusion is between Working Hours Issue and Worker Issue. Perfect results for tiny classes are unstable because Minor Delay and Quality Issue each contain only four real rows.

Model 2 — overall order-risk level (v3)

Binary Low Risk / High Risk classification with broader schedule features

Candidate	Accuracy	Macro-F1	Weighted-F1	ROC-AUC	Decision
Dummy baseline	5.85%	5.53%	0.65%	50.00%	
Logistic Regression	99.67%	98.52%	99.67%	100.00%	
Random Forest	99.83%	99.25%	99.83%	99.99%	Selected
Gradient Boosting	99.83%	99.25%	99.83%	99.82%	



SELECTED MODEL

Random Forest

Highest Macro-F1; tie broken by ROC-AUC

ACCURACY

99.83%

597 of 598 rows classified correctly

MACRO-F1

99.25%

Class-balanced summary

FALSE LOW-RISK ROWS

1

Confusion matrix: [[35, 0], [1, 562]]

Critical validity limitation

Only one independent real Low Risk order is available, even though it contributes 35 daily rows. Therefore the near-perfect row-level score is research evidence, not proof of generalisation to new low-risk orders. The artifact is explicitly marked production_approved: false.

Early warning — next 3 production days

Supported subtype models use calibrated probabilities and target-specific thresholds

Machine breakdown

Random Forest

Accuracy	**86.21%**
Macro-F1	**84.05%**
Positive-class F1	**78.18%**

Decision threshold: 0.3239

Quality-limit issue

Logistic Regression

Accuracy	**66.67%**
Macro-F1	**51.18%**
Positive-class F1	**78.68%**

Decision threshold: 0.7263

Output/schedule risk

Logistic Regression

Accuracy	**79.31%**
Macro-F1	**63.75%**
Positive-class F1	**40.00%**

Decision threshold: 0.1951

Target coverage on eligible stable rows

Target	Positive / negative rows	Positive / negative orders	Grouped modelling
Machine breakdown	47 / 127	11 / 11	Ready
Quality-limit issue	134 / 40	11 / 11	Ready
Output/schedule risk	25 / 149	10 / 11	Ready

Calibration evidence

Calibration reduced Brier score and expected calibration error for all three targets. Threshold metrics shown above come from nested grouped validation, not the final training fit.

Scope boundary

A general any-emergency warning and a worker-shortage warning are not supported: the eligible stable-day data lacks enough negative rows and independent negative orders.

Source: reports/component3_early_warning_step5b_calibrated/evaluation.json

Model selection rationale

Each selected model is compared against a non-predictive baseline and alternative algorithms

Prediction task	Candidates compared	Selected	Evidence
Current risk type	Dummy, Logistic Regression, Random Forest, Extra Trees	Random Forest	Macro-F1 90.08%
Overall order risk v3	Dummy, Logistic Regression, Random Forest, Gradient Boosting	Random Forest	Macro-F1 99.25%*
Machine warning	Dummy, Logistic Regression, Random Forest, XGBoost	Random Forest	Nested calibrated Macro-F1 84.05%
Quality warning	Dummy, Logistic Regression, Random Forest, XGBoost	Logistic Regression	Nested calibrated Macro-F1 51.18%
Output/schedule warning	Dummy, Logistic Regression, Random Forest, XGBoost	Logistic Regression	Nested calibrated Macro-F1 63.75%

Why Macro-F1 matters

Accuracy can look high when one class dominates. Macro-F1 gives each class equal weight, so model selection better reflects minority classes such as incidents and low-risk cases.

Why keep the dummy baseline

The dummy classifier shows what can be achieved by always choosing the most frequent class. A useful model should improve materially over this non-predictive reference.

Interpretation notes

- Model 1: Random Forest improves Macro-F1 from 6.97% (dummy) to 90.08%.
- Model 2 v3: the score is high, but only one real Low Risk order exists; the asterisk marks this important limitation.
- Early-warning model names come from candidate comparison; displayed warning metrics use calibrated nested thresholds.
- Selected .pkl/.joblib artifacts package preprocessing and the fitted estimator for application inference.

Sources: Component 3 evaluation JSON and model-comparison CSV evidence

Recovery engine validation

Rule-based recovery planning is evaluated separately from the ML classifiers

CURRENT-OUTPUT WAPE

13.63%

398 next-day cases

RECOMMENDED-CAPACITY WAPE

12.54%

276 observational cases

COMPLETION-DATE MAE

0.218 d

97.10% within 3 days

DEADLINE ACCURACY

69.10%

Single observed outcome class

Evaluation	Cases	Absolute error	Squared error	Tolerance result
Current output → next-day baseline	398	MAE 46.312	RMSE 59.800	Within 10% 36.20%
Recommended capacity comparison	276	MAE 43.806	RMSE 61.367	Within 10% 52.90%
Completion-date forecast	280	MAE 0.218 days	—	Within 3 days 97.10%

Correct interpretation

The engine calculates feasible recovery options from current output, remaining quantity, days available, and optional workforce/machine limits. It is deterministic business logic, not a trained ML model.

Evidence limitation

Historical data records recommendations and outcomes, but not which action was actually applied. Capacity comparisons are observational and must not be described as causal improvements.

All 405 observed deadline cases met the deadline, so missed-deadline discrimination cannot be measured.

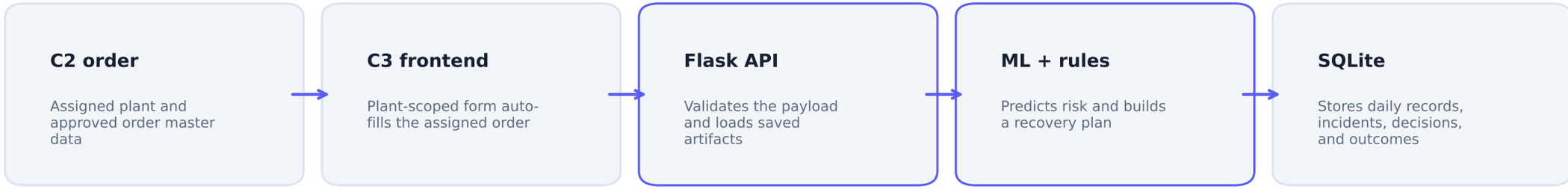
Calendar assumptions: Monday–Friday production; factory/public holidays are not represented.

Source: reports/component3_recovery_validation/evaluation.json

Research implementation details

How Component 3 is trained, served, integrated, secured, and persisted

IMPLEMENTATION ARCHITECTURE



Responsibility	Implementation
Training	train_component3_v2.py; train_component3_model2_v3.py; train_component3_early_warning.py
Feature engineering	components/component3_features.py; component3_order_risk_features.py; component3_early_warning_data.py
Runtime inference	components/component3.py; components/component3_early_warning_inference.py
Recovery planning	components/component3_recovery.py — deterministic, explainable business rules
Daily persistence	components/component3_monitoring.py → instance/component3_tracking.db
Incident workflow	components/component3_tracking.py → decisions, status transitions, and actual outcomes
Frontend	frontends/src/app/(dashboard)/component3/monitoring and recovery-history

Backend and model serving

Flask registers Component 3 under /api/component3. The default v3 loader reads the risk-type v2 pickle, order-risk v3 pickle, and three calibrated early-warning joblib artifacts lazily.

Frontend integration

Next.js/TypeScript calls http://127.0.0.1:5001/api/component3 by default. The monitoring screen sends authenticated prediction, monitoring-record, and incident requests.

Access control

Manager and PlantManager routes require authentication. JWT plant_id scopes Plant Manager data; the API never trusts a browser-supplied plant query parameter.

GitHub project and reproducibility

Verified repository location, branch, project structure, and repeatable commands

GITHUB REPOSITORY

https://github.com/IT22548146/R26-IT-002

Active development branch: component3

The URL above is the configured origin of the current local project. Repository visibility depends on GitHub access permissions.

Project path	Purpose
components/	Component APIs, feature engineering, monitoring, recovery, validation
models/	Saved .pkl and .joblib inference artifacts plus metadata
data/	Component 3 historical and controlled augmented workbooks
reports/	JSON/CSV evaluation evidence and this generated PDF
frontends/	Integrated Next.js portal and Component 3 user interfaces
tests/	Backend, integration, validation, access-scope, and workflow tests

REPRODUCIBILITY COMMANDS

```
python train_component3_v2.py
python train_component3_model2_v3.py
python train_component3_early_warning.py
python scripts/generate_component3_ml_report.py
python main.py                          # backend :5001
cd frontends && npm run dev             # frontend :3000
```

Runtime configuration

Backend uses Flask on port 5001; frontend uses Next.js on port 3000. NEXT_PUBLIC_COMPONENT3_API_URL can override the C3 API base. COMPONENT3_MODEL_VERSION defaults to v3. COMPONENT3_TRACKING_DB can override the C3 SQLite path.

Repository hygiene

Keep .env, API keys, SMTP passwords, local databases, virtual environments, node_modules, and generated build caches out of Git. Commit source code, approved model artifacts, dataset documentation, evaluation evidence, and reproducible scripts.

Before sharing: rotate any previously exposed credentials and provide .env.example with placeholder values only.

Artifacts, conclusions, and reporting limits

Reproducible evidence map for code review, viva, and final documentation

Evidence type	Repository path
Historical dataset	data/component3_final_preprocessed_dataset.xlsx
Risk-type training data	data/component3_training_augmented.xlsx
Overall-risk v3 data	data/component3_training_balanced_research_v2.xlsx
Current risk model	models/c3_model1_risk_type_v2.pkl
Overall order-risk model	models/c3_model2_order_risk_v3.pkl
Early-warning models	models/c3_early_warning_*_v2.joblib
Model 1 evaluation	reports/component3_v2/evaluation.json
Model 2 v3 evaluation	reports/component3_model2_v3/evaluation.json
Warning evaluation	reports/component3_early_warning_step5b_calibrated/evaluation.json
Recovery validation	reports/component3_recovery_validation/evaluation.json

What is demonstrated

End-to-end model training, grouped evaluation, artifact persistence, calibrated subtype warnings, API inference, and recovery-plan generation.

What is not claimed

No claim of production approval, causal recovery benefit, general emergency warning, or worker-shortage future warning from the current evidence.

Next evidence needed

More independent orders, more minority-class examples, several real Low Risk orders, applied-action logs, and missed-deadline outcomes.

CONCLUSION

Component 3 has a coherent research pipeline and performs substantially above dummy baselines on the available order-grouped evidence. The results support a working academic prototype, while the stated data limitations justify retaining research-only status until broader independent validation is collected.